

Basics of Knee Arthroscopy

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Few areas in orthopaedic surgery have grown as rapidly as knee arthroscopy. Arthroscopy often can be performed more quickly and with increased accuracy, lower complication rates, decreased hospitalization time, and shorter recovery periods, compared with many more open operative techniques. The effective use of arthroscopy is based on the understanding of the benefits and indications for arthroscopy, as well as its limitations.

The knee was the first joint to be examined arthroscopically, and many of the fundamental principles of arthroscopy were developed for the knee.¹ The first knee arthroscopy was performed in Europe and was advanced significantly by Japanese surgeons (Takagi and Watanabe).¹ Applications continue to expand, and the future scope of arthroscopic applications is limited only by the imagination of the arthroscopist.

PREOPERATIVE EVALUATION

Indications for knee arthroscopy continue to expand at a rapid rate. Each patient's unique anatomy must be considered before initiating arthroscopy. Systematic evaluation of the entire knee includes a thorough physical examination and history. Additional studies including radiographs and advanced imaging should be reviewed, and proper documentation must be performed.² Pre-operative consultation with appropriate medical specialties and an anesthesiologist help reduce perioperative complications.³ Postoperative prophylaxis for deep vein thrombosis (DVT) should be considered in at-risk patients. Local, regional, and general anesthetic considerations should be reviewed with the patient and the anesthesia team.

INDICATIONS

Arthroscopy has diverse application in various forms of knee disease. Diagnostic arthroscopy helps confirm suspected knee injuries.⁴ An arthroscopic synovectomy can be useful for synovial biopsies to aid in the diagnosis of rheumatologic disorders, to remove diseased synovium and loose bodies, and to resect synovial folds or plicae. Arthroscopic treatment of septic arthritis of the knee has increased in frequency. Treatment of meniscal disease is perhaps the most common application of arthroscopy. Meniscal tears and repairs account for about half of knee injuries that require surgery. Osteochondral lesions commonly are addressed arthroscopically. Microfracture, autologous

chondrocyte implantation, and osteochondral plug transfers are also performed arthroscopically.

Injuries to the cruciate ligaments can be diagnosed easily with arthroscopy and subsequently treated. Arthroscopic-assisted reconstruction of these ligaments is one of the most common orthopaedic procedures today.⁵ Other procedures that sometimes are aided with arthroscopy include tibial plateau fracture reduction, reduction and fixation of tibial eminence fractures, loose body removal, anterior fat pad débridement, and lateral release for patellar malalignment.³

Contraindications for knee arthroscopy must be considered as well. One such consideration includes local skin infections over the portal sites. In addition, alternative treatments should be considered for patients who have too high a risk for surgery and those who are not expected to be compliant with postoperative rehabilitation.⁶

POSITIONING

Two different forms of positioning are commonly used for knee arthroscopy. The patient can be positioned supine on the operating table, and a lateral post can be used for countertraction. Alternatively, the operative leg can be positioned in a commercially available leg holder (Fig. 92.1). The operative leg is allowed to hang freely over the end of the operating table, and the opposite leg is positioned in a well-padded leg holder, taking care not to compress the peroneal nerve.

PORTAL PLACEMENT

Landmarks, including the inferior pole of the patella and the joint line, are marked. Portal incisions are typically vertical, 1 cm in length, and made with a no. 11 blade while the knee is flexed.⁴ A spinal needle can be used for localization of the anteromedial portal (Fig. 92.2). Portal placement is key to successful knee arthroscopy. Standard arthroscopic portals for knee arthroscopy have traditionally included a superomedial or superolateral portal for fluid inflow and outflow, and inferomedial and inferolateral portals positioned just above the joint line on both sides of the patellar tendon for arthroscopy and instrumentation (Fig. 92.3).⁷ Typically the inferolateral portal is used for arthroscopic visualization and the inferomedial portal is used for instrumentation, although alternating instrumentation between the medial and lateral portal is often necessary to reach certain structures

Abstract

Indications for knee arthroscopy continue to expand at a rapid rate. In this chapter, we discuss the basic principles of knee arthroscopy.

Keywords

knee
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basics
technique
principles

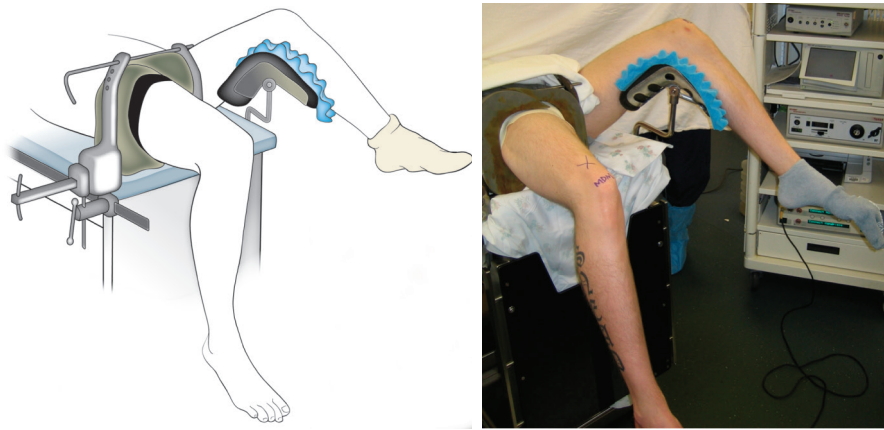


Fig. 92.1 The patient is positioned supine with use of a leg holder on the lateral post. (From Miller MD, Chhabra AB, Safran MR. *Primer of Arthroscopy*. Philadelphia: Elsevier; 2010.)

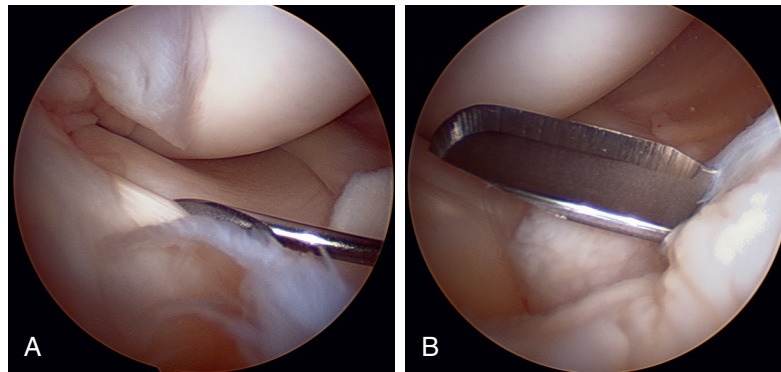


Fig. 92.2 Anteromedial portal localization. (A) The arthroscope is turned toward the anteromedial capsule. A spinal needle is inserted and positioned until it is just above the meniscus and its trajectory is satisfactory for intra-articular work. (B) The spinal needle is carefully removed, and a no. 15 or no. 11 blade is inserted blade-side up.

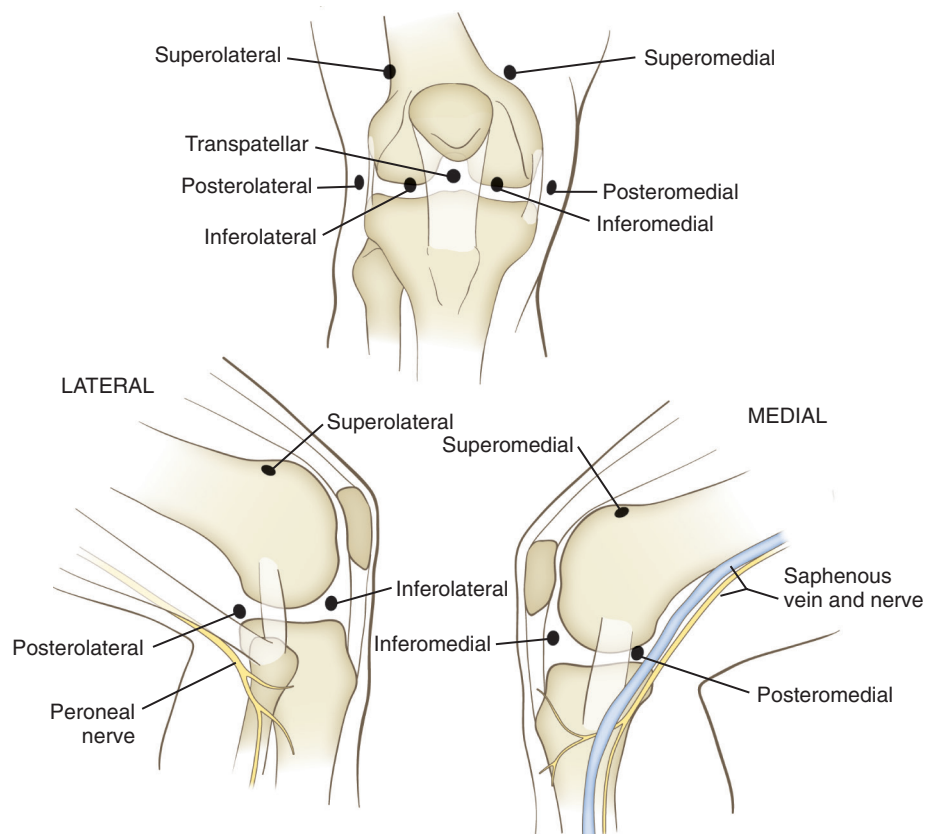


Fig. 92.3 Portal placement for knee arthroscopy. (From Miller MD, Chhabra AB, Safran MR. *Primer of Arthroscopy*. Philadelphia: Elsevier; 2010.)

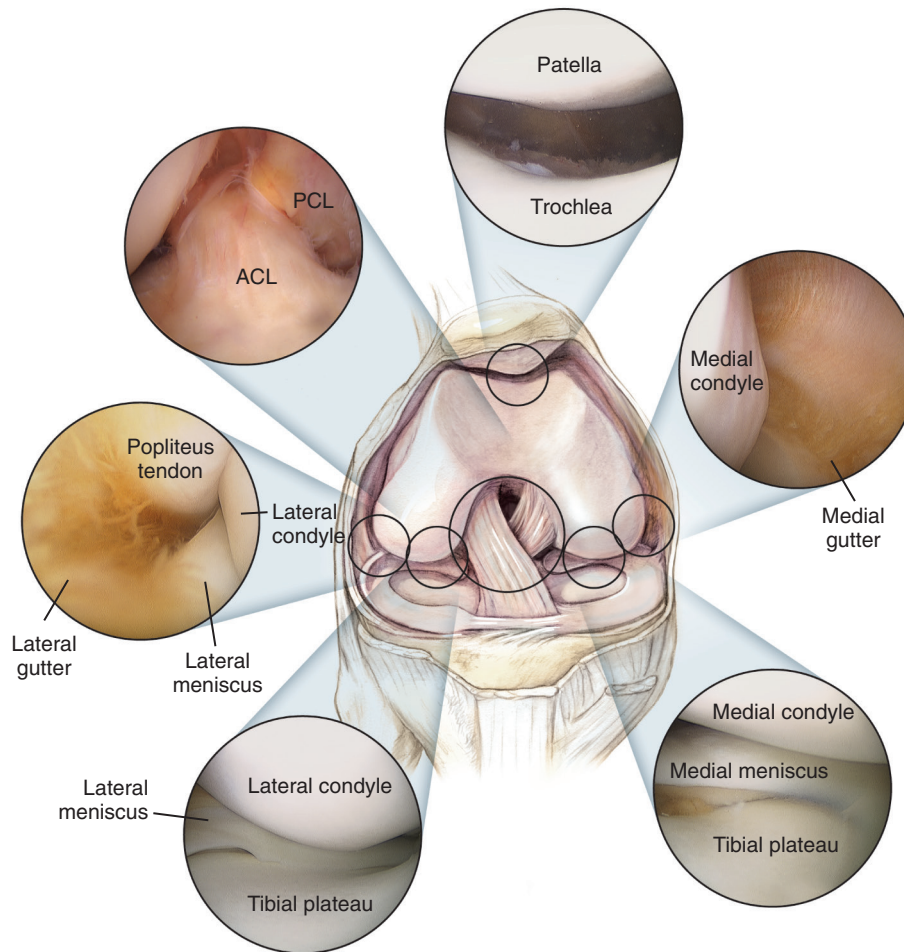


Fig. 92.4 Visualization during diagnostic arthroscopy. ACL, Anterior cruciate ligament; PCL, posterior cruciate ligament. (From Miller MD, Chhabra AB, Safran MR. *Primer of Arthroscopy*. Philadelphia: Elsevier; 2010.)

(Fig. 92.4). Newer arthroscopic fluid control systems have now made the use of superior outflow portals optional. The use of a far proximal superior portal can still be helpful for the visualization of patellar tracking (see Fig. 92.3).

Accessory portals for the knee include the posteromedial, posterolateral, far medial and lateral, and proximal superomedial portals. The posteromedial portal is often helpful for visualizing the posterior cruciate ligament and the posterior horn of the medial meniscus (Fig. 92.5).⁸ The posterolateral portal, located just posterior to the lateral collateral ligament between the iliotibial band and the biceps tendon, sometimes is helpful, but extreme care should be taken to ensure that the portal is anterior to the biceps tendon to avoid injury to the peroneal nerve (Fig. 92.6). An accessory medial portal has been developed for obtaining access to the appropriate angle for anatomic femoral tunnel placement in anterior cruciate ligament (ACL) surgery.⁹ Other portals include the midpatellar portal, far medial and lateral portals (which are sometimes helpful for instrument placement in hard-to-reach areas), and the proximal superomedial portal, located 4 cm proximal to and in line with the medial edge of the patella (for assessment of patellar tracking).

DIAGNOSTIC ARTHROSCOPY

As with any joint, systematic examination of the knee is appropriate. Before positioning the patient, a complete examination is conducted after induction of anesthesia to assess instability in all planes. An arthroscopic cannula is placed in the superomedial or superolateral portal for inflow and outflow (although the use of these superior portals is now optional with many of the new pump systems), and the obturator and sheath are introduced into the inferolateral portal after incision with the knee flexed at 60 to 90 degrees, angled toward the notch. As the knee is brought into extension, the obturator and sheath are advanced into the suprapatellar pouch and the obturator is replaced by a camera for visualization. The anteromedial portal can be made at the outset of the case or created under visualization with the spinal needle (see Fig. 92.2). Although many examination sequences are possible, it is important to visualize the suprapatellar pouch, patellofemoral joint (Fig. 92.7), medial and lateral gutters, medial and lateral compartments (meniscus and articular cartilage), and intercondylar notch (cruciate ligaments) in all patients.⁴ Surgeons differ with regard to fat pad excision for visualization and therapeutic purposes.

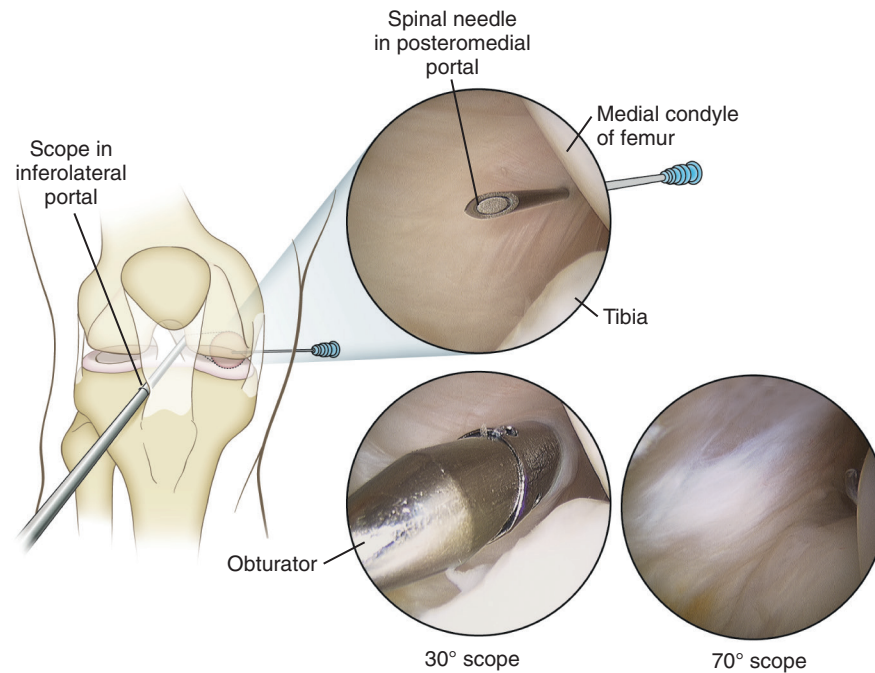


Fig. 92.5 View of the intercondylar notch from the posteromedial portal. (From Miller MD, Chhabra AB, Safran MR. *Primer of Arthroscopy*. Philadelphia: Elsevier; 2010.)

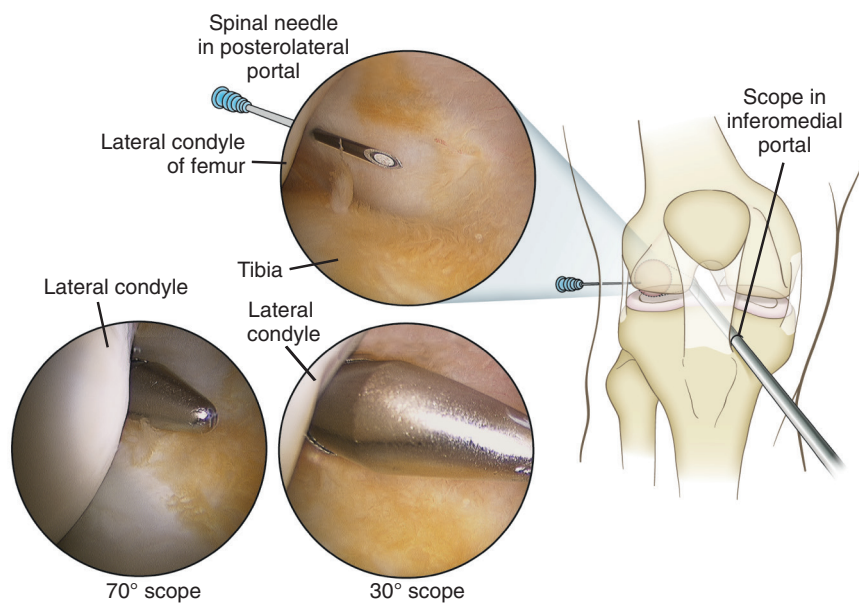


Fig. 92.6 View of the intercondylar notch from the posterolateral portal. (From Miller MD, Chhabra AB, Safran MR. *Primer of Arthroscopy*. Philadelphia: Saunders; 2010.)

The patellofemoral joint is inspected and articulation is examined, including the patella facets and trochlea. Engagement should be full at 40 degrees. The gutters are examined for loose bodies. The knee is flexed and the scope is brought down into the intercondylar notch. The ACL is inspected by directing the scope to view it laterally and by probing the ligament (Fig. 92.8). The ACL is composed of the two separate bundles that are often not distinct but can occasionally be recognized. The posterior cruciate ligament is also evaluated, although often only the femoral side is examined, with the remainder hidden by the ACL.¹⁰ The

medial compartment is then visualized with valgus stress and extension using an assistant or with the surgeon resting the ankle on his or her hip. The foot can be externally rotated to improve access. The meniscus and articular surfaces are examined. Once all lesions are characterized, the lateral compartment is visualized in the same fashion in the figure-of-four position.

Accessory viewing portals are established as necessary if other areas need to be evaluated. A posteromedial portal can be helpful whenever medial meniscus pathology is suspected but is unable to be identified from the anterior portals. This portal is established

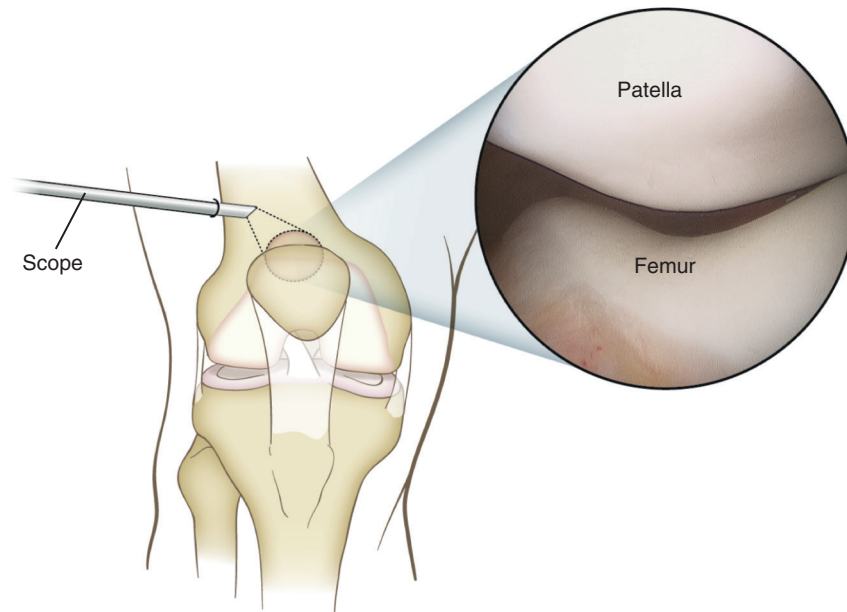


Fig. 92.7 Visualization of the patellofemoral joint. (From Miller MD, Chhabra AB, Safran MR. *Primer of Arthroscopy*. Philadelphia: Elsevier, 2010.)

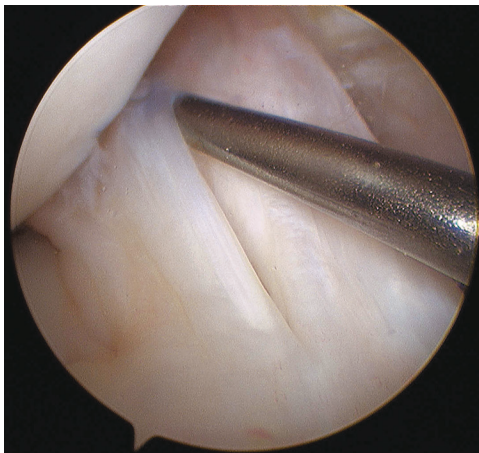


Fig. 92.8 Arthroscopic appearance of the two bundles of the anterior cruciate ligament. (From Miller MD, Chhabra AB, Safran MR. *Primer of Arthroscopy*. Philadelphia: Elsevier, 2010.)

by introducing the arthroscopic cannula into the back of the knee by directing it from anterior to posterior on the notch side of the medial femoral condyle. Care must be taken to avoid the saphenous nerve and vein. A spinal needle is used to establish the position of the portal. Next, a small incision is made in the skin only, followed by spreading with a blunt instrument down to the capsule. Once the arthroscope is in the posterior aspect of the knee, the posterior horn of the medial meniscus can be visualized. Use of a 70-degree scope may be helpful. After a complete evaluation of the joint is performed, all surgical pathology is addressed accordingly.

INSTRUMENTATION

A 30-degree arthroscope is most commonly used, although a 70-degree scope may be helpful in the posterior corners.⁸ An

arthroscopic probe can be used and provides a sense of touch to the arthroscope. Instruments angled upward, including biters, are best for the medial compartment, whereas straight instruments often work best in the lateral compartment. Arthroscopic shavers are available in both large and small sizes and should be chosen on the basis of the dimensions of the compartment.⁵

POSTOPERATIVE CARE

Upon completion of the procedure, the fluid is evacuated from the joint and the instruments are removed. Typically nonabsorbable monofilament is used to close the portals and is removed 10 to 14 days after surgery. In some centers, portals are routinely left open, because it is believed that open portals permit easier fluid extravasation and reduce postoperative swelling and pain.¹¹ A local anesthetic can be injected at the end of the surgery into the skin only. Although the weight-bearing status may differ depending on the surgery performed, most patients require crutches for some period of time. Postoperative pain medicines are typically prescribed for a short period for most patients undergoing knee arthroscopy.

COMPLICATIONS

Postoperative complications can include infection, arthrofibrosis, nerve and vessel injury, iatrogenic cartilage injury, compartment syndrome, collateral ligament injury, and DVT or pulmonary embolism.^{6,12} Perhaps the most common complication is inadvertent damage to the intra-articular structures.¹² This risk is inversely proportional to the surgeon's experience and the care with which the surgeon performs the procedure. Proper portal placement, use of a gentle technique, and attention to detail are crucial. The exact prevalence and long-term sequelae of iatrogenic cartilage lesions are unknown, but studies involving second-look

arthroscopy and animal models have shown that this is a true risk and that the lesions do not tend to fill with time. Nerve or vessel injury can result from improper placement of the portals.⁷ For this reason, a thorough knowledge of local anatomy is necessary before performing arthroscopy.

Tourniquet paresthesia can be reduced with use of a wide cuff and by limiting tourniquet time whenever possible. Positioning devices (e.g., leg holders) can have a tourniquet effect, even when the tourniquet is not inflated. Fluid extravasation has been reported, although the incidence of this complication can be reduced with careful placement of inflow cannulas and proper use of inflow pumps. Synovial fistula formation is a rare complication of arthroscopy and is usually remedied by 7 to 10 days of immobilization and, occasionally, delayed closure. Infection is an extremely rare complication of arthroscopy with less than a 1% occurrence rate and is usually the result of a break in sterile technique. High-dose antibiotics and arthroscopic irrigation and débridement may be indicated in severe cases. Hemarthrosis can occur as well, although it can be avoided with meticulous technique and care after procedures such as a lateral release.⁵

Arthrofibrosis is unusual after knee arthroscopy, although it can occur with procedures such as ACL reconstruction and multiligamentous knee reconstructions. Attainment of full motion preoperatively and early motion postoperatively are key to avoiding such an outcome.

An unusual but potentially life-threatening complication of knee arthroscopy is the development of a postoperative DVT or pulmonary embolism. Although the risk for these events is low, careful screening of risk factors must be performed to identify patients who require routine prophylaxis.^{3,6} These risk factors include immobilization, increased body mass index, personal or family history of clotting disorder, and advancing age.

SUMMARY

Knee arthroscopy is effective in treating a number of intra-articular knee pathologies. Proper evaluation and management are key to successful patient outcomes. Results vary based on the underlying etiology, but they do appear to help patients in the symptomatic management of intra-articular disorders.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Hoogeslag RA, Brouwer RW, van Raay JJ. The value of tourniquet use for visibility during arthroscopy of the knee: a double-blind, randomized controlled trial. *Arthroscopy*. 2010;26(suppl 9):S67–S72.

Level of Evidence:

I

Summary:

In this randomized controlled trial, significantly improved visibility during routine knee arthroscopy was demonstrated with use of a tourniquet.

Citation:

Jameson SS, Downen D, James P, et al. The burden of arthroscopy of the knee: a contemporary analysis of data from the English NHS. *J Bone Joint Surg Br*. 2011;93B(10):1327–1333.

Level of Evidence:

II

Summary:

In this article a prospective, large-volume registry analysis of knee arthroscopies occurring in the United Kingdom National Health Service is provided. The overall rate of complications was less than 1%.

Citation:

Sikand M, Murtaza A, Desai VV. Healing of arthroscopic portals: a randomised trial comparing three methods of portal closure. *Acta Orthop Belg*. 2006;72(5):583–586.

Level of Evidence:

II

Summary:

In this article a prospective randomized controlled trial comparing suture closure of arthroscopy portals with Steri-Strip application or no closure whatsoever is described. Swelling was significantly lower in the two groups without suture closure. Patient satisfaction with regard to final scar appearance was higher in the two groups without suture closure.

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